

Short Questions 4

02/15/2026

Your answers should be formal with mathematical formulas and algorithms, as necessary. Type your responses and upload a PDF file. If you want to add a drawing, you can include a nicely hand-drawn figure or a figure drawn with a computer tool. Learn how to type formulas in Word or LaTeX. Use single spaced typing. Preferably, keep your answers to a page.

1. (4 points) Describe how a reinforcement learning agent, one that is not forced to randomly perform every $\langle state, action \rangle$ pair at the beginning of an episode with non-zero probability, can attempt to explore every $\langle state, action \rangle$ pair during training. In other words, how can we write a non-Exploring-Starts algorithms that learns values? Write out any formulas needed.
2. (2 points) Differentiate between on-policy and off-policy reinforcement learning algorithms.
3. (4 points) When an agent learns $v_\pi(s)$ in a target policy π from a behavior policy b , it needs to transfer G_t values (or Returns) from policy b to policy π . What formula can an agent use to do so? Write the formula and discuss it briefly.